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Computing Systems

**Live Session on Computer System
Architecture – Memory and I/O**

Dr. Biju K Raveendran

Course Coverage of Computing Systems

- 3 Credits Course with 13 Weeks of Content
- Prof. Biju Raveendran [Week 1 to Week 5] and Prof. Shan Sundar Balasubramaniam [Week 6 to Week 13]
- W1: Getting started with Programming [Prof. Biju]
- W2: Computer System Architecture – Memory and I/O [Prof. Biju]
- W3: Introduction to an Operating System [Prof. Biju]
- W4: Shell Scripting [Prof. Biju] [Assignment #1 [15%] – Based on W3 and W4 with a 2 weeks Deadline to submit] [W4 end to W6 end]
- W5: Program Execution in a System [Prof. Biju] [Assignment #2 [15%] – Based on W5 – with a 3 weeks Deadline after the Assignment #1 submission is done] [W6 end to W9 end]

Course Coverage of Computing Systems

- W6: Virtual Environment and Isolation [Prof.Shan] [Assignment #1 ends. Assignment #2 starts with a deadline of 3 weeks.]
- W7: Source code Repositories and Versioning Systems [Prof.Shan] [Graded Quiz #1, 10%, Coverage: Week 4, Week 5, Week 6 and Week7] [30 min]
- W8: Profiling and Debugging [Prof.Shan]
- W9: Introduction to GPUs [Prof.Shan] [Assignment #2 ends.]
- W10: Program Execution and Performance Tuning in GPUs [Prof.Shan] [Graded Quiz #2, 10%, Coverage: Week 8, Week 9 and Week10] [30 min]
- W11: Introduction to Distributed Systems [Prof.Shan]
- W12: Introduction to Cloud Computing [Prof.Shan]
- W13: Introduction to Mobile / Embedded Computing [Prof.Shan] [Graded Quiz #3, 10%, Coverage: Week 11, Week 12 and Week13] [30 min]

Course Evaluations of Computing Systems



- Assignment #1 [15%] [Coverage Week3 and Week4] [Timing: Week4 to Week6]
 - Assignment #2 [15%] [Coverage Week 5] [Timing: Week6 to Week9]
 - Graded Quiz #1 [10%] [Coverage Week4 to Week7] [Timing: Week7]
 - Graded Quiz #2 [10%] [Coverage Week8 to Week10] [Timing: Week10]
 - Graded Quiz #3 [10%] [Coverage Week11 to Week13] [Timing: Week13]
 - Final Examination 40%
-

Locality of Reference

Locality of Reference

- Importance of Locality in Programming

How to access data from Memory?

How to access data from Disk?

- Memory Mapped I/O

Locality of Reference

Spatial locality

```
int i, j, A[1024][1024];  
    for (i=0; i<1024; i++)  
    {  
        for(j=0; j<1024; j++)  
        {  
            A[i][j] = i+j;  
        }  
    }
```

Locality of Reference

Temporal locality

```
int i, j, A[1024];  
    A[0]=0; A[1]=1;  
    for (i=2; i<1024; i++)  
    {  
        A[i] = A[i-1] + A[i-2];  
    }
```


Locality of Reference

Spatial and Temporal localities

```
int i, j, A[1024][1024], x=5;
  for (i=0; i<1024; i++)
  {
      A[i][0]=x;
      for(j=1; j<1024; j++)
      {
          A[i][j] = A[i][(j-1)] +x;
      }
  }
```

Semi-Conductor Memories

Why Semi-conductor memories

Properties

- Random Access
- Access time depends on
 - Make of the memory (Static RAM Vs Dynamic RAM)
 - Size (Small L1 Cache to High Main memory – DRAM)
 - Where it is placed (L1, L2 Main memory)

Categorization and its requirement

- RAM: SRAM and DRAM
- ROM: MROM (Mask ROM), PROM (Programmable ROM), EPROM (Erasable PROM), EEPROM (Electrically EPROM), UVPROM (Ultraviolet)

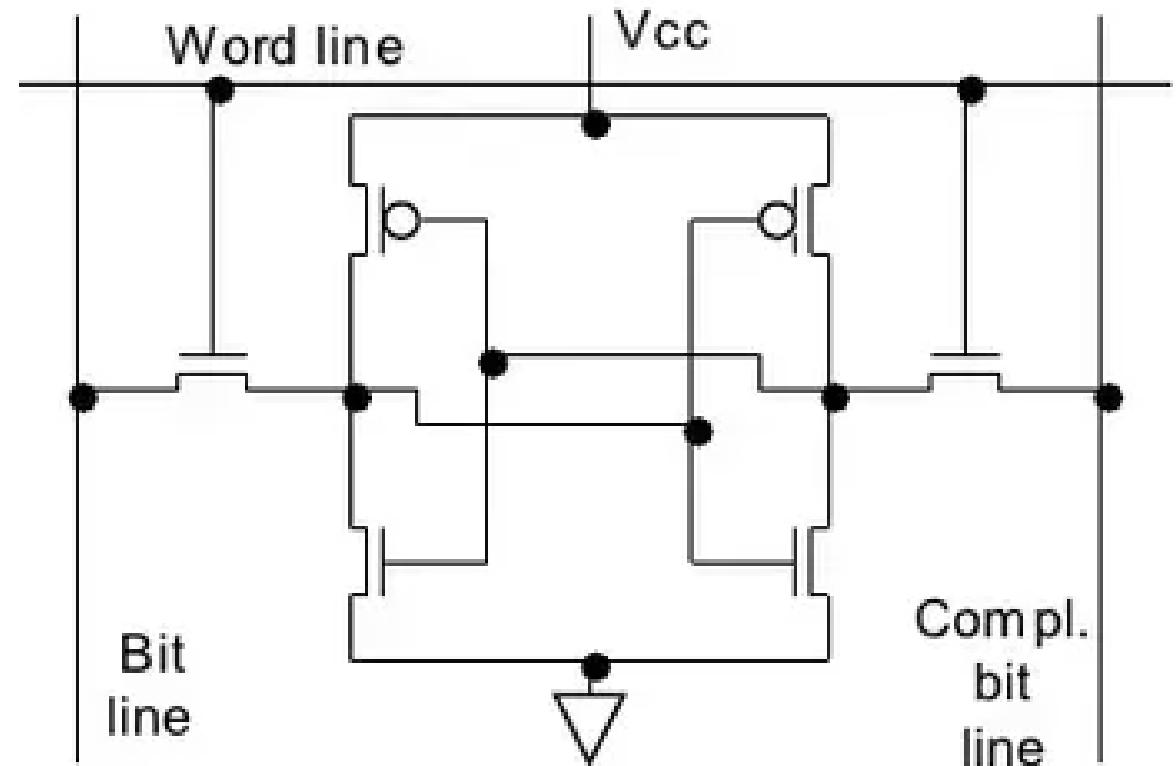
Static Random Access Memory

SRAM structure

- 6 transistors
- Very high switching speed
- L1 / L2 / L3

Properties

- Inclusive
- Transparent to OS
- Dedicated hardware controller



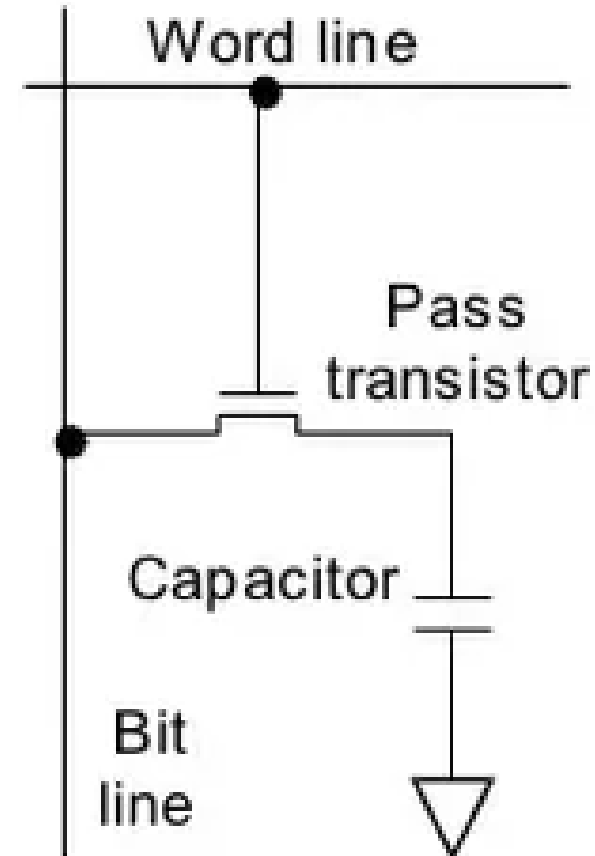
Dynamic Random Access Memory

DRAM structure

- Capacitor and transistors
- Low switching speed
- Requires refresh circuit to retain data
- Main Memory

Properties

- Inclusive
- Controlled by OS
- Software controlled



Secondary Memory

Access of Data from I/O devices

Dual Mode protection

Persistent data storage

Accessing a File

Secondary Memory

Block Devices and Character Devices

Character Devices

```
int var1;  
scanf("%d",&var1);
```

Block Devices

```
int var1;  
FILE *fp;  
fp=fopen("CS/temp.txt","r");  
fscanf(fp,"%d",&var1);
```

Buffer Cache

Hard Disk



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Hard disk structure

- Track, Sector
- Zoned Hard disks

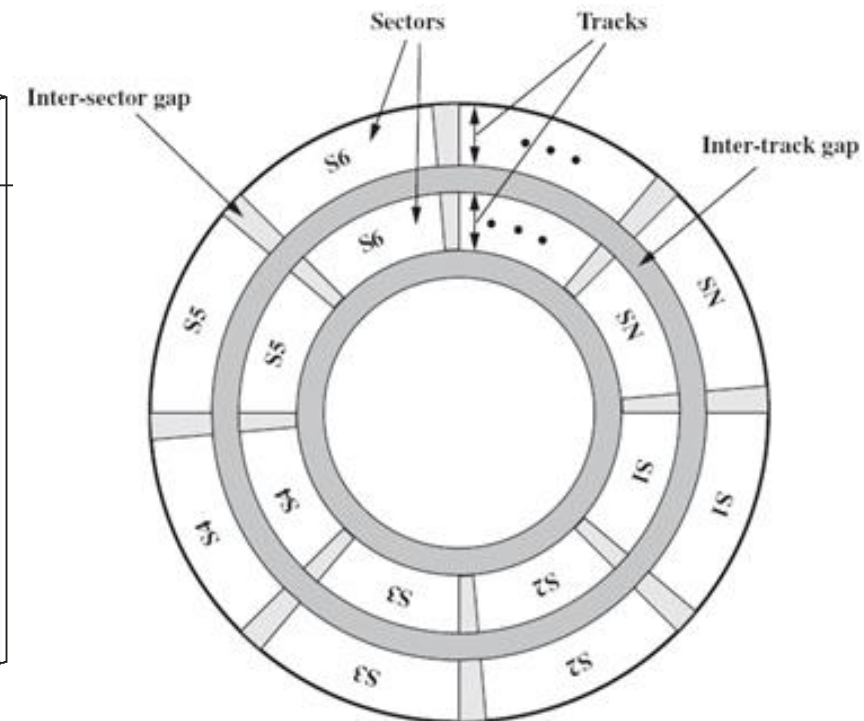
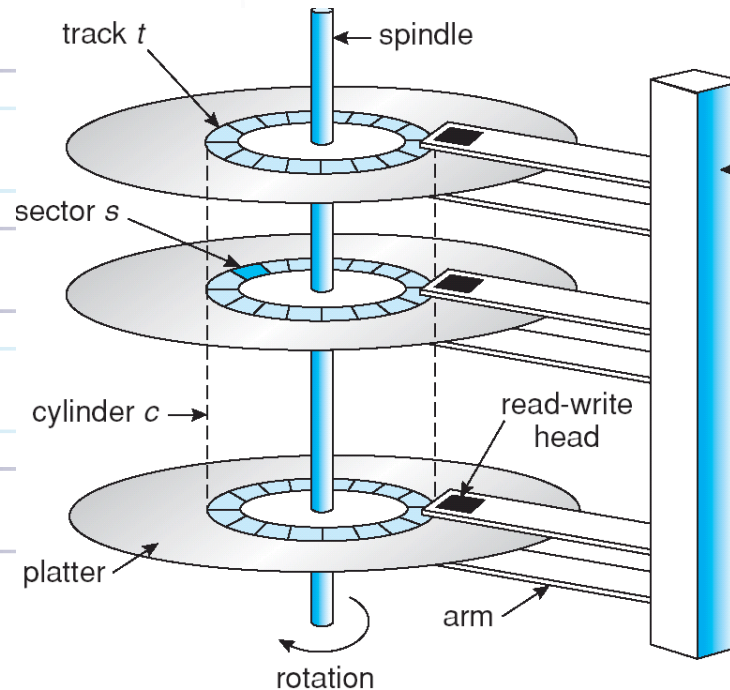
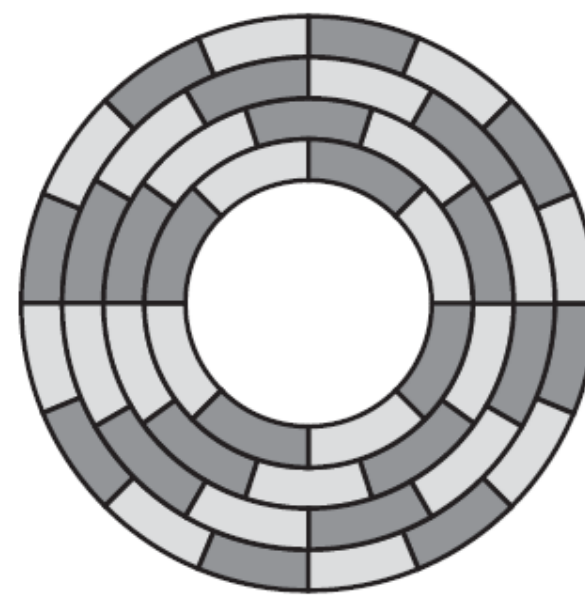
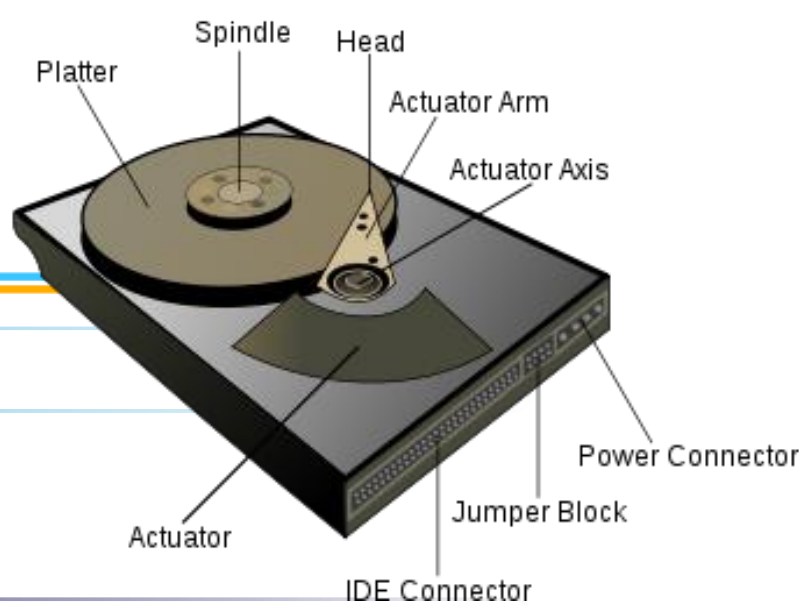
Disk cache in Hard disk controller

Buffer Cache in Main Memory

Maintaining consistency

Row partition Vs Cooked Partition

Virtual Memory Support



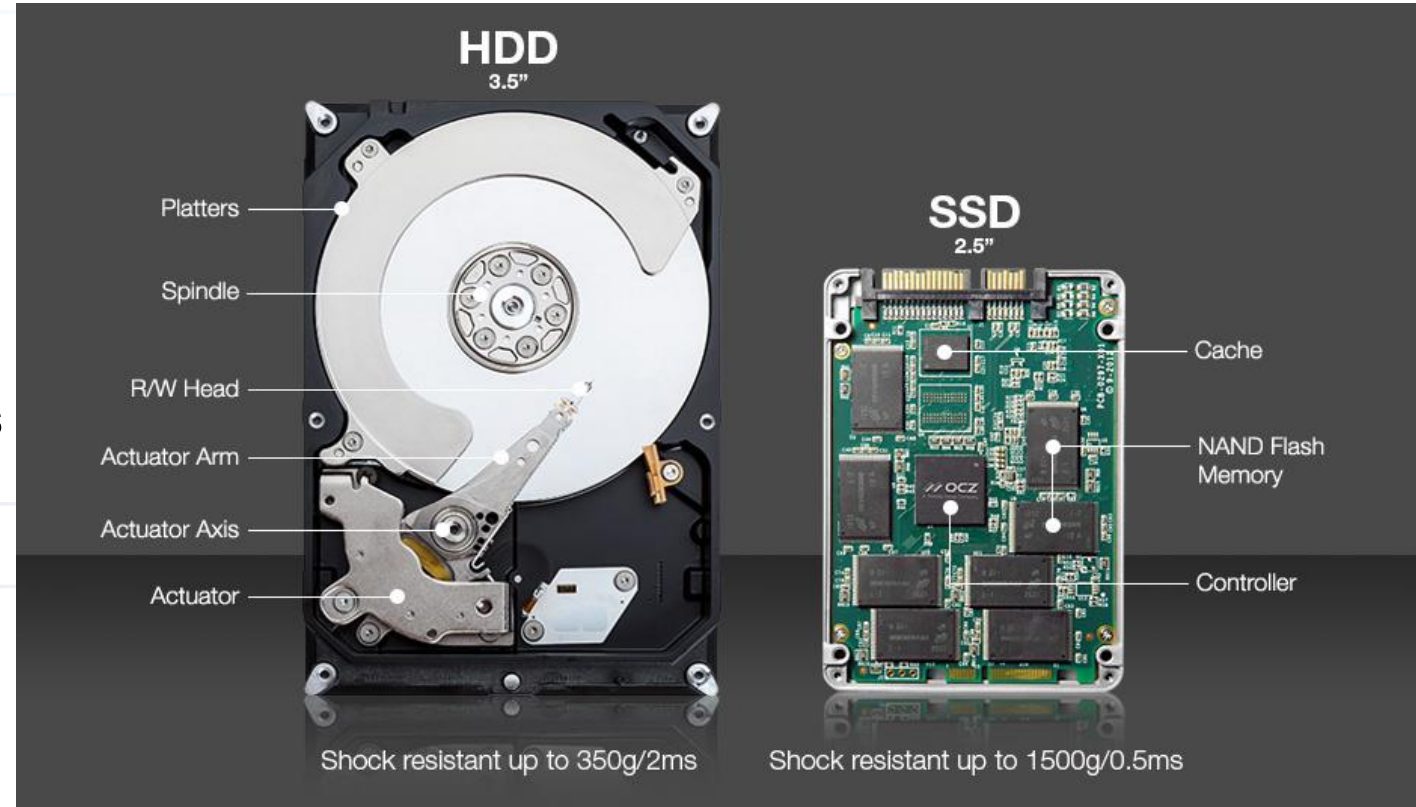
SSD

Access

- Block access
- Random Access
- Reads are fast
- Require Erase before Write
- Sectors as Erase units – Group of blocks

Performance

- Constant Access Time
- Writes very slow compared to Read
- Wear levelling required
- Accumulate writes in main memory to improve performance



SSD

Solid State Drives – Primary Vs Secondary Storage

- **NOR flash**
 - Supports bit-wide modifications
 - Execute in Place (XIP) memory – Can be used as ROM
 - Low write speed
 - Does not require ECC
- **NAND flash**
 - High density, Large capacity, Low cost
 - Requires erase unit

	NAND	NOR
Cost per bit	Low	High
Capacity	High	Low
Read Speed	Medium	*High
Write Speed	High	Low
File Storage Use	Yes	No
Code Execution	Hard	Easy
Stand by Power	Medium	Low
Erase Cycles	High	Low
Application	File Storage	Code Execution
Active Power	Low	Medium
Standby Power	Medium	Low

I/O Devices

Input Devices

- Keyboard, Joy stick, bar code reader, Scanner

Output Devices

- Head phones, Printer, Speaker

I/O Devices

- Disk/SSD, Touch screen, Pen drive, Webcam, Interactive Projector

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